AJAN MUTHURAJ

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RESEARCH AND COMPETITION HIGHLIGHTS

- **Multi-Robot Frontier-Based Exploration under Uncertainty with SLAM and Temporal Map Fusion** · Co-author, ICRA 2026 XPLORE workshop poster.
- **2nd among Indian teams**, Swarm Rescue Challenge 2025–26 · Built a ten-drone search-and-rescue system with the IIT Madras team.
- **6th place**, e-Yantra Robotics Competition 2024–25, Logistic coBot · Validated UR5/eBot package handling at IIT Bombay and was selected for eYSIP.
- **Top 10**, e-Yantra Robotics Competition 2023–24, GeoGuide · Built a working vision-guided rover that routed to events in priority order.

EDUCATION

PROGRAM	INSTITUTION	CGPA / %	YEAR
Mechanical Engineering (B.Tech)	Indian Institute of Technology Madras	**8.52/10**	**2026**

RELEVANT SKILLS

Robotics: ROS 2, Nav2, MoveIt 2, SLAM, motion planning, manipulation, computer vision, sensor fusion, state estimation and feedback control.
Simulation and Learning: NVIDIA Isaac Sim, Gazebo, reinforcement learning (RL), PyTorch, MATLAB, RViz and familiarity with Isaac Lab.
Software and Tools: C++, Python, Linux, Git, CMake, Docker Compose, OpenCV, software testing, and continuous integration with GitHub Actions.
Mechatronics: SolidWorks, Fusion 360, Arduino, ESP32, sensor integration and motor control. **Languages:** English, Hindi, Tamil, Japanese (basic).

PROFESSIONAL EXPERIENCE

Systems Engineer **DLTX, Japan** **Oct 2026 - Present**	• Reverse-engineered **Lotus Notes** applications from XML exports to identify form structure, data fields and legacy screen behavior. • Mapped legacy form definitions into reusable **React components** and built a browser interface from the supplied XML exports. • Studied **Java/Spring Boot**, data persistence, Docker and Kubernetes to understand how web applications are built and deployed.
Humanoid Robot **IIT Roorkee · Intern** **Jun - Jul 2025**	• Converted SolidWorks models to URDF and integrated **Nav2**, SLAM and differential-drive control for humanoid navigation in Gazebo. • Linked **RGB-D perception** and 3D object localization to MoveIt 2 and OctoMap so the robot could approach and grasp around obstacles. • Recorded **96.6% grasp success** across 30 simulated trials and documented navigation and manipulation in separate demonstrations.

PROJECTS

EXTRACURRICULAR PROJECTS

Isaac Sim UAV Autonomy	• Bridged Isaac Sim physics to **ArduPilot software-in-the-loop (SITL)** and integrated RGB, IMU, range and 3D LiDAR in ROS 2 Jazzy. • Built **3D frontier exploration** with RTAB-Map and OctoMap for collision avoidance, online replanning and automatic return-to-home. • Ran **nine drones** in Isaac Sim and published Docker Compose tooling, Python unit tests and GitHub Actions CI. • Fixed stale physics samples in the ArduPilot bridge and measured **5 cm peak horizontal error** over 61.7 s of simulated hover.
Time-Varying Reach-Avoid RL **B.Tech Thesis**	• Added time to the **reach–avoid Bellman state** so a policy could react to moving obstacles and target zones in custom environments. • Trained a **PyTorch Double DQN** with experience replay and target-network updates to navigate under time-varying safety constraints. • Measured **94.4% precision** among states predicted safe in the two-obstacle simulation and documented the result in the thesis poster.
Swarm Rescue Challenge 2026	• Contributed adaptive frontier selection and **temporal occupancy-grid fusion** to coordinate exploration across ten simulated UAVs. • Contributed to **FastSLAM/ICP localization** and integrated hazard detection and collision recovery for GPS-denied search. • Built for **single-operator supervision**, using shared maps to track fleet exploration during simulated search-and-rescue missions.
Logistic coBot **e-Yantra, IIT Bombay**	• Integrated a **UR5 arm and eBot rover** in ROS 2 to pick up, sort and deliver packages across a shared logistics workspace. • Localized packages with **ArUco** for MoveIt 2 Servo pick-and-place and fused rover measurements with an EKF for ultrasonic docking. • Validated the complete **docking-to-pickup sequence** in Gazebo and on physical UR5 and eBot hardware at IIT Bombay.
Cubitect **Electronics Club** **Project Lead**	• Led a modular robotics project exploring self-reconfiguring cubes that use reaction wheels and controlled toppling to move. • Built vision-based tracking with **A* path planning** and Hungarian assignment to match individual cubes with target positions. • Developed the actuation and coordination approach and worked on a ROS 2 and Gazebo environment to test multi-cube behavior.
Swarm Navigation **Inter-IIT Tech Meet**	• Developed a ROS 2 and Gazebo simulation workflow for groups of mobile robots navigating a dynamic warehouse-like environment. • Combined LiDAR obstacle clustering with **DBSCAN**, map merging and TD3 policy training for multi-robot navigation. • Published recordings, checkpoints and four- and five-robot launch configurations with the IIT Madras Inter-IIT robotics team.
GeoGuide **e-Yantra, IIT Bombay**	• Built a physical mobile robot that used an overhead camera to localize arena events and navigate to them in priority order. • Combined **ResNet-50** event classification and Dijkstra routing with ArUco tracking and QGIS georeferencing for the arena. • Sent route commands to an ESP32 rover over Wi-Fi/TCP and used infrared sensors for on-robot line following during execution.

ACADEMIC PROJECTS

Decentralized Swarm Navigation	• Extended **decentralized flocking** in MATLAB with shared occupancy-grid memory so robots could navigate an unknown maze. • Implemented **deadlock recovery** to escape traps and delayed self-reinforcement to damp oscillations under noisy LiDAR sensing.
Adaptive RRT* Motion Planning	• Compared Adaptive RRT*, RRT*, A*, Adaptive A* and Lifelong Planning A* in simulated environments with dynamic obstacles. • Implemented **incremental replanning** that keeps valid route segments and searches from a safe waypoint when obstacles move. • Evaluated planners on path efficiency, robustness and computation, with source code and a static replanning diagram linked.
Dual-Axis Solar Panel Tracker	• Designed a dual-axis panel mechanism in Fusion 360 and used light-dependent resistors to sense the direction of incident light. • Implemented **PID feedback** for two servos and added an OLED display for live voltage and status monitoring.

POSITIONS OF RESPONSIBILITY

Electronics Club **Core**	• Managed **recruitment, funding and inventory** for the IIT Madras Electronics Club's student projects and technical workshops. • Taught a hands-on **parallel processing** workshop at Shaastra and helped run an IoT smartwatch hackathon from design to build.
Electronics & Robotics Guild **Head**	• Led **electronics and robotics training**, selected members and assembled teams for Inter-IIT technical competitions. • Represented IIT Madras in the **Bharat Forge swarm robotics challenge** and coordinated resources for student engineering teams.

EXTRA-CURRICULAR

- Captained Alakananda Hostel to **two IIT Madras football titles** in hostel competition and represented the institute at intercollegiate football tournaments.